

1a	Velocity v_i represents the object's velocity at one instant and v_f its velocity at a later instant. In the vector triangle, the change in velocity $\Delta v (=v_f - v_i)$, is due to a change in the velocity's direction, while the magnitude of v_i and v_f are the same. The direction of this change in velocity Δv is towards the centre of the circle, resulting in a centripetal acceleration perpendicular to velocity. Hence, the object is accelerating. • Vector diagram showing that velocity changes direction, with brief description. • State that force or acceleration is perpendicular to velocity, and relate a change in direction of velocity to acceleration.
1bi	• gravitational force by earth on man (vertically downwards) • contact force by wall on man (diagonal to the left. Vertical component of contact force to be equal in magnitude to weight). Accept: Contact force represented by Friction and Normal Reaction in correct direction and magnitude
1bii	The <u>upward frictional force is equal and opposite to the downward gravitational force</u> acting on the rider. Hence, the rider would not fall off. The <u>normal contact force by the wall on the rider provides for the centripetal force</u> necessary for him to move together with the barrel in a circular path.
1biii	acceleration = $r\omega^2 = r(2\pi f)^2 = 2.5(2\pi)^2(33/60)^2$ $= 29.9 \text{ m s}^{-2}$
1biv	Work done = <u>0 J</u> because the <u>net force acting on the man</u> (normal contact force) is <u>always perpendicular to the velocity (or direction of motion) of the man</u> at any instant. As the instantaneous velocity is in the same direction of instantaneous displacement, hence, work done which is the product of force and the displacement in the direction of the force would be zero. Or

	From the work-energy theorem, since there is no change in kinetic energy of the rider, there is no work done on the rider.
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2a	The increase in internal energy ΔU of a system is equal to the sum of the heat
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